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Exploring speaker fluency with phonologically annotated ICE corpora

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ABSTRACT: This study demonstrates how quantitative characteristics of speaker fluency can be measured in the phonologically annotated and time-aligned corpora ICE-Nigeria and ICE-Scotland, which belong to the ‘new generation’ of ICE corpora. Some files from the categories broadcast talk and unscripted speeches in ICE Nigeria and ICE Scotland were phonemically annotated and analysed, and for each speaker the mean length of run (= average number of words per utterance) and articulation rate (= mean number of phonemes per total articulation time) were calculated. Mixed effects regression models with SPEAKER as a random factor show that mean length of run in scripted but not in unscripted speech is significantly longer in Scottish than in Nigerian English ($p < 0.05$). Articulation rate depends, in both varieties, mainly on the length of utterances, with longer utterances being spoken faster. When this is taken into account, Scottish speakers have a higher articulation rate than Nigerian speakers in scripted ($p < 0.05$) but not in unscripted speech. These results show how phonological and time-aligned annotations can enrich ICE corpora, and how they allow comparisons of varieties of English that go beyond studies of syntactic and lexical variation.

INTRODUCTION

Since its conception more than 25 years ago (Greenbaum 1991), the ICE project has been an enormous success with the number of ICE corpora growing steadily and the number of empirical studies using ICE data ever on the increase (see e.g. Biermeier 2008, Lange 2012, Deuber 2014 and the studies in Hundt & Gut 2012). While clearly the original corpus design

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still stands strong, technical advances in the past two and a half decades have called into question some of the formal stipulations for the corpus compilation process suggested by the project's founding fathers and mothers, and some empirical studies have revealed potential for improvement in the areas of data presentation and accessibility. As a consequence, a new generation of ICE corpora has been developed (e.g. Wunder et al. 2010) that addresses these issues and suggests new approaches in terms of data format, transcription process, time-alignment of transcriptions and metadata management (see section 2). These changes offer a wide range of new research opportunities, including the automatic calculation of quantitative features of speaker fluency, as will be exemplified in section 3 with a pilot study based on data drawn from ICE Nigeria and ICE Scotland. In section 4, a discussion and outlook is given.

A NEW GENERATION OF ICE CORPORA

The four major weaknesses of the original ICE corpus compilation process and guidelines that have emerged in the past years include

- the choice of the data format SGML,
- the abundance of errors in the manual transcriptions and annotations,
- the lack of time-alignment of the transcriptions with the sound files,
- the incoherent handling of metadata

When the ICE corpus data format was conceived in the 1990s, an SGML-based format was chosen as a standard data format at the time. By now, however, SGML has mostly fallen out of use so that texts coded with this mark-up scheme cannot easily be searched or otherwise processed with modern tools anymore. Instead, XML (Bray et al. 1998) has become the standard data format for linguistic data (e.g. Romary & Witt 2014). Compared to SGML it has three advantages: it can be validated with XML validators, tools are available for its transformation into other data formats and it can be searched with languages such as XQuery.

Second, the manual transcriptions of the corpus files with the tagset specified in the ICE Markup Manuals (Nelson 2002) have proven rather difficult and error-prone for transcribers. As can be seen in Figure 1, which presents an extract from the file S1A-007 in ICE Jamaica, transcribers have to produce highly complex and for human readers uninterpretable strings of symbols such as “<.,>” (line 1) and “<.,>” (line 5), which stand for a long and a short pause respectively, and “<[> </[> ... <{> </{>”, which stands for overlapping speech, in their transcriptions, which make mistakes very likely.

```
<$A>At the centre well uhm<.,> pretty much as I said <#>Here at the center <}><-
><.>every</.></-> as for my personal point of view <=>everything</=></}> is
fine<#>Work is great a few glitches here and there but I mean nothing to really
complain much about <#>Uhm I like what I do as I told you <}><->I'm</->
<=>I'm</=></}>pretty much<.,> into computing <}><->that's</-> <=>that's</=></}>
what I like <#>I really like uhm programming and graphic art and uhm just the whole
multimedia aspects of computing <}><->that's</-> <=>that's</=></}> really what I like
about it all<.,> right and I uhm I was here <unclear>word</unclear> and I really think I
want to pursue a career along that line like into heavy desktop publishing
```

Figure 1. Example of manual SGML markup of file SA1-007 in ICE Jamaica.

For empirical studies that require the further automatic processing of the ICE corpora such as part-of-speech tagging and chunking (e.g. Schneider & Hundt 2012), laborious corrections of all transcription errors thus first need to be carried out. Third, the transcription guidelines for the older generation of ICE corpora do not suggest time-aligning the transcriptions to the sound files of the spoken part so that, with the exception of ICE-GB that provides access to the sound files albeit no time-aligned transcriptions, the older generation ICE corpora cannot be used for phonological research at all.

A fourth drawback of many of the older generation ICE corpora lies in their incoherent handling of metadata. While some corpora provide extensive metadata on the speakers/writers of the texts contained in the corpus, others provide none at all. But even if metadata is provided, the format in which it is given ranges from individual text files for each file (e.g. ICE India) to databases, or Word or Excel files comprising all information (e.g. ICE Philippines). Any empirical studies that include speaker information such as their gender and age therefore require manual extraction of this information in a very time-consuming process (e.g. see Höhn 2012). The new generation ICE corpora have addressed each of these four problems and have come up with the following solutions that increase the potential of new ICE subcorpora but at the same time ensure full comparability with the older ones.

The new generation ICE corpora are created with the software *Pacx* (*Platform for Annotated Corpora in XML*; Gut 2011; www.pacx.sf.net), a tool that was developed for the compilation of ICE Nigeria (Wunder et al. 2010) and that produces a corpus with an XML-based data format. It combines linguistic data management, annotation, searching and distribution and constitutes an extended Eclipse platform (www.eclipse.org) enriched by a set of tools that includes the XML editor *Vex* for the transcription of the written part, *ELAN* (Sloetjes 2014; www.lat-mpi.eu/tools/elan) for the annotation of audio and video files, and Subversive, a client for the version control system Subversion that supports collaborative work. The annotation of the written data is based on an XML template that is automatically created by *Pacx* whenever a new raw data file is added to the corpus. In contrast to the traditional error-prone manual annotation of the earlier ICE corpora, the annotation of the new generation ICE corpora with *Pacx* does not require transcribers to type any of the SGML tags specified in the Markup Manual for Written Texts (Nelson 2002). Instead, the transcriber marks the word or phrase that is to be annotated and selects an appropriate tag (e.g. 'italics') from a predefined list that is shown when a word to be annotated is selected (see Figure 2). The integrated XML editor *Vex* automatically creates the relevant XML tags for each

annotation label chosen by the transcriber. Different annotations are visualised with different text formats for the transcriber; for example words annotated as a heading will appear in bold (see Figure 2). *Pacx* saves all annotations as a well-formed error-free XML document. As the manual for ICE Nigeria (Gut 2014b) shows, care was taken to have equivalents for all types of markup, e.g. boldface and paragraphs, that are suggested in the ICE Markup Manual for Written Texts so that older and newer ICE corpora can be easily compared.

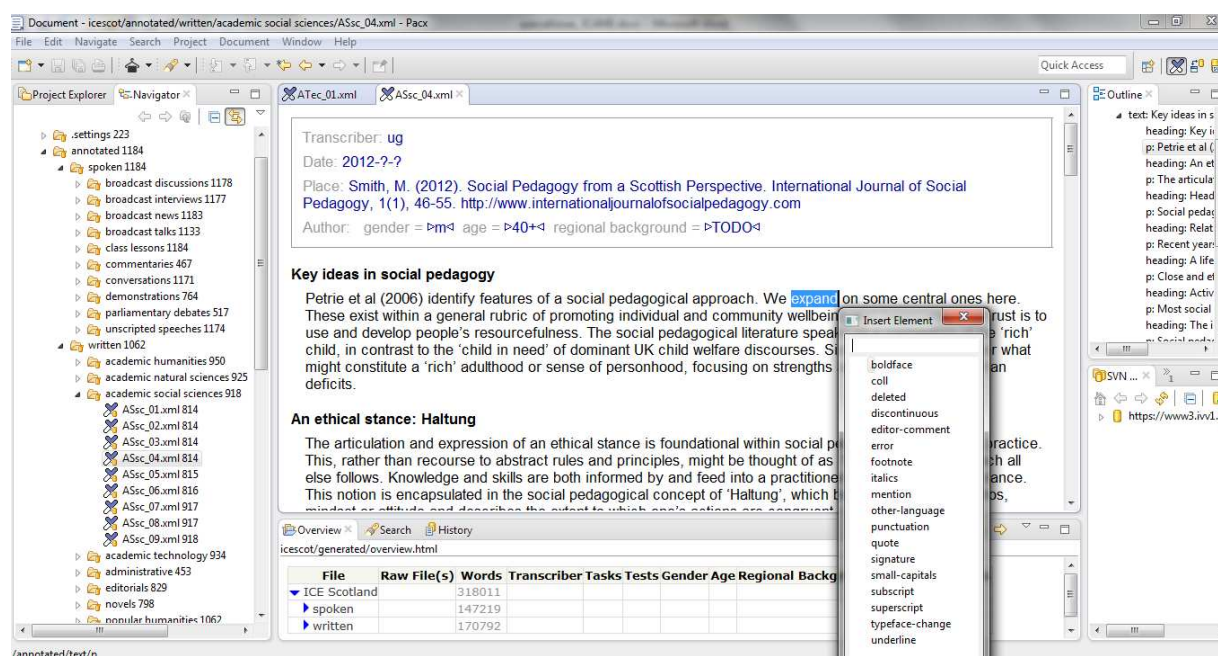


Figure 2. Annotation of the written part of new generation ICE corpora using *Pacx*.

For the transcription of the spoken part, *Pacx* creates a template for *ELAN*, a tool that allows time-aligned transcription of audio and video data. In the process of text-to-tone alignment the transcriptions carried out by the transcribers are linked with the corresponding parts of the recording. Technically speaking, the boundaries of all transcribed elements are defined by time stamps, which means that information about the exact beginning and end of each annotated element is available in the corresponding XML-based file that is created by *ELAN*.

Figure 3 illustrates the transcription of a spoken file in ICE Scotland that is carried out with the software *ELAN*. Beneath the speech wave form, the transcription is represented on three different tiers for each speaker. On the top tier, the beginning and end of each utterance, defined in the project as an intonation phrase (which equals roughly an interpausal stretch of speech), are marked and utterances are transcribed orthographically. Analogous to annotating written files with *Vex*, adding additional annotation of the spoken part is very simple and does not require any typing. Instead, transcribers choose annotation labels from predefined lists. *ELAN* allows the creation of a customized list of annotations in the form of a so-called Controlled Vocabulary, from which the transcriber chooses the correct annotation to insert on the relevant tier, which are shown on the middle tier in Figure 3. For example, individual words can be marked as a repair and anonymised words can be indicated. On the bottom tier transcribers can add values like 'first name' to specify the type of anonymised word further.

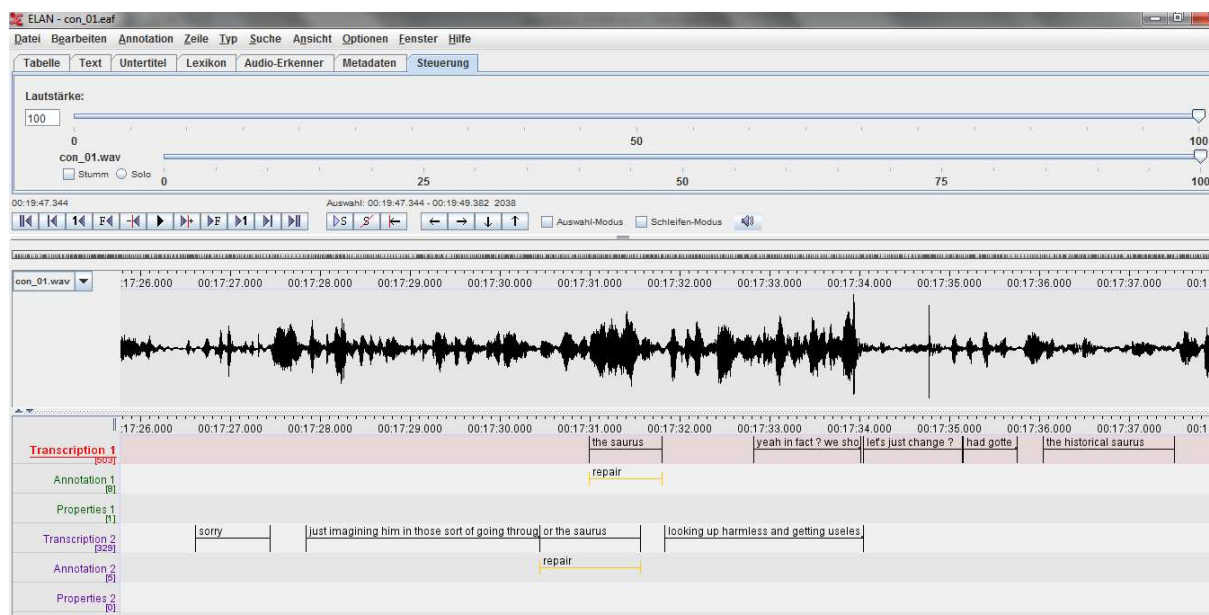


Figure 3. Transcription of a private conversation in ICE Scotland with *ELAN*.

Using *Pacx* for the transcription of the spoken part of ICE corpora speeds up the transcription process enormously. Most of the mark-ups specified in the ICE Markup Manual become

superfluous: transcribers do not need to guess the length of pauses anymore and indicate whether they are perceived as long or short; instead, by the placement of utterance boundaries, the exact length of each intervening pause can be calculated automatically. Likewise, speaker overlap does not need to be transcribed anymore as the placement of utterance boundaries indicates precisely overlapping speech (see Figure 3). Furthermore, like for the transcription of the written data, none of the mark-ups need to be typed by hand but are created automatically by *ELAN* in an XML-based format. At the same time, the other annotations specified in the ICE Markup Manuals guidelines such as unclear words, mentions and indigenous words are used to make the new ICE corpora fully comparable to the older ones. Apart from offering a whole new range of possibilities for analysing phonological properties of spoken language with new generation ICE corpora, the time-aligned annotation illustrated in Figure 3 assists the annotation process crucially by providing direct access from each transcribed element to the corresponding part of the original audio or video file. By clicking on any transcribed element, this will be played back. By the same token, transcriptions can effortlessly be checked and, if necessary, altered by later corpus users.

Further annotations created by external tools can easily be added to new generation ICE corpora, for example part-of-speech (POS) tagging that consists of the identification and annotation of the word class of each transcribed lexical item. The written part of the ICE Nigeria was tagged for parts-of-speech by Prof. Stella Neumann at the University of Aachen using the TreeTagger (Schmid 1994) and the Penn Treebank tagset (<http://www.cis.uni-muenchen.de/~schmid/tools/TreeTagger/data/Penn-Treebank-Tagset.pdf>, Santorini 1991), which for example allowed the analysis of progressive and perfect forms used in Nigerian English (Gut & Fuchs 2013, Fuchs & Gut 2015, Werner & Fuchs 2016). Moreover, in ICE Scotland a phonemic forced alignment system known as the Munich Automatic Segmentation System (MAUS, Schiel 2004) is employed for the automatic creation of time-aligned

phonemic transcriptions (this type of phonemic annotation is called ‘forced’ because it always produces an annotation, no matter the quality of the input).

Figure 4 shows all steps involved in this process (see Appendix for a step-by-step guide). We start out with an orthographic annotation created manually in *ELAN*, with time-aligned utterances (see Figure 3). This file, saved in *ELAN*’s XML-conform eaf format, includes information on (a) the start and end time of utterances and (b) the words uttered by the speakers. This file is then exported from *ELAN* as a Praat TextGrid file. We have also found it useful at this stage to remove all utterances that include words marked as unclear, since phonemic annotation cannot be expected to be accurate in this case.

Next, Grapheme to Phoneme conversion (G2P, Reichel 2012) transforms the orthographic transcription into a phonemic transcription, while retaining information about the start and end time of utterances. This step is optional, but greatly reduces the complexity of the forced alignment (and thus increases its accuracy), since the forced aligner already starts with information on where roughly in the recording each word occurs.

The phonemic forced alignment proper then takes as input the phonemic transcription and the sound recording, and produces a time-aligned phonemic transcription. This is achieved with the help of an acoustic model of the language (or dialect) that has been trained on manually annotated data. The acoustic model contains information on the acoustic structure of the phonemes of the target language. Based on this information, the forced aligner determines which stretches of the sound file best match the phonemes derived from the orthographic transcription. The end result is another Praat TextGrid file, which provides information on the start and end points of the phonemes included in this sound file. Optionally, word and utterance tiers can be included. If the previous step (Grapheme to Phoneme conversion with time alignment of utterances) is skipped, the forced aligner takes as input a text file with an orthographic transcription instead of a TextGrid file.

How accurate automatic phonemic annotation is, and whether manual correction is necessary, depends on a number of factors. Since even trained phoneticians do not reach 100 % agreement in their annotations, the best we can hope an automatic system to achieve is to come within the range of human inter-annotator agreement. For some tasks this may be possible. In order to compute, for example, duration-based metrics of speech rhythm (see Fuchs 2014, 2016), forced phonemic alignment can reach the range of human inter-annotator agreement (Wiget et al. 2010). On the other hand, for the fluency analysis presented in the second half of this paper, we felt it necessary to manually correct the automatic annotation because it is not known how accurate forced phonemic alignment is for these purposes.

The accuracy of forced phonemic alignment is also influenced by how closely the accent of the speaker matches the acoustic model (or, by extension, the data it was trained on). Even within a single variety there might be considerable geographical or sociolinguistic variation. More generally, acoustic models are not (yet) available for many World Englishes. For example, the Scottish and Nigerian data used in the second half of this paper was processed with forced alignment based on a British English acoustic model because Scottish and Nigerian English acoustic models are not yet available, and we are currently working on a detailed evaluation of how this affects the accuracy of forced phonemic alignment. Despite these limitations, most users find that forced phonemic alignment greatly speeds up the task of producing phonemic annotations. Previous studies based on ICE corpora have led to a tremendous body of work on syntactic, morphological and lexical variation completed in the previous two decades. With phonemically annotated ICE corpora of the new generation, we can now build on this success and widen the perspective to study phonological variation in World Englishes. A wide range of investigations will be possible on the segmental (e.g. vowel quality and quantity, rhotics, aspiration of voiceless plosives) and suprasegmental (speech rhythm, speech rate) level.

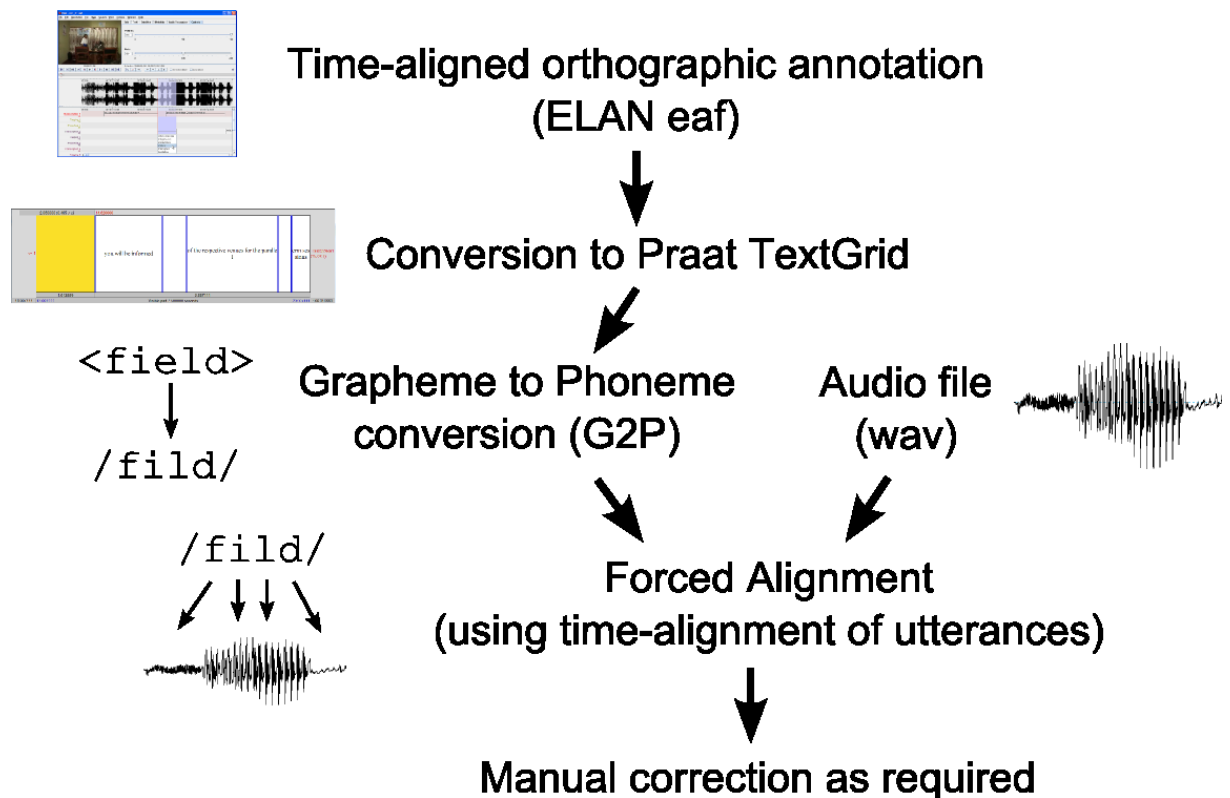


Figure 4. Step-by-step guide for phonemic annotation of corpus data.

With regard to the fourth weakness identified in older ICE corpora above, *Pacx* allows the integration of metadata into each XML file. For the spoken data, the metadata such as speaker age, gender and further languages can be added to the appropriate tier, and general metadata concerning the time and place of recording is added to the first tier. For the written data, the transcriber can enter all relevant metadata such as the place and time of writing of the raw file, transcriber ID and gender, age and regional background of the author/s in the header of the automatically created XML template (see Figure 2).

Last but not least, ICE corpus compilation with *Pacx* allows collaborative work with participants from anywhere in the world. The corpus data is stored on a central server that provides version control. Each collaborator has a local copy of the corpus data downloaded from the server onto his or her computer. Changes to the corpus data are carried out and are first saved locally. In a separate step, they are transferred to the central server. The collaborators stay synchronized by regularly updating their local copy to the latest version of

the corpus data from the server. If transcribers do not remember whether they committed their changes to the server, they can use the Synchronize function, in which all incoming and outgoing changes that have occurred since the last update are made visible, as well as potential conflicts that might have arisen. To solve these conflicts, a transcriber can either discard local changes by overwriting the local files with the newest version of the corpus or choose to commit the local files as a newer version. Last but not least, when a transcriber commits some change by mistake, the version control system offers a safety net in saving the history of each transcription file and allowing the recovery of older versions.

Since one of the aims of the ICE project is to compare national varieties of English, one might ask in how far differences in the compilation of ICE corpora of the original and the new generation affect their comparability. Given that most of the annotations (such as pauses or indigenous words) are equivalent across both corpus generations but more reliable in the new generation and checked by several transcribers, we would like to argue that corpora of the new ICE generation are a closer approximation to the ideal of an error-free corpus. Problems with comparability might perhaps arise in one particular case, i.e. in conversations with several participants, when analysts want to compare turn-taking across corpora. In the traditional markup system, this information is represented orthographically through different types of brackets. In the new generation of ICE corpora, on the other hand, patterns of turn-taking and overlap are stored in the ELAN annotations together with the relevant timestamps. However, orthographic representations equivalent to those of the traditional corpora can be derived automatically from the corpora of the new generation.

To date, there is one completed ICE corpus of the new generation (ICE Nigeria[‡]; Wunder et al. 2010) and another three corpora are under construction that will become new generation ICE corpora: ICE Bahamas, compiled at the University of Munich (Hackert 2010), ICE Malta, compiled at the University of Bamberg (Hilbert & Krug 2010), and ICE Scotland,

[‡] ICE Nigeria is available at <<http://sourceforge.net/projects/ice-nigeria/>>

compiled at the University of Münster (Schützler, Gut & Fuchs to appear), which currently comprises more than 460,000 words. The second part of this article will show how phonemic annotation of second generation ICE corpora can be used to study fluency in varieties of English spoken as a native and as a second language.

MEASURING SPEAKER FLUENCY IN ICE NIGERIA AND ICE SCOTLAND

Fluency is a term that is used by a wide range of researchers and practitioners including linguists, psycholinguists, researchers in second language acquisition, language teachers and language testers. Accordingly, conceptualisations of fluency vary widely and comprise definitions such as the general mastery of and proficiency in a language, the fluidity, continuity or smoothness of spoken language production, the speed of the cognitive processes involved in the planning and delivery of spoken language and the degree of command and mastery of conversational conventions achieved by a speaker in a foreign language (e.g. Schmidt 1991, Brumfit 2000, Fiksdal 2000).

Many empirical studies in the area of speaker fluency are carried out within the psycholinguistic approach, where fluency is proposed to reflect the efficacy of the cognitive processes involved in speech planning and production. The psycholinguistic conceptualisation of speech production is based on Levelt's (1989) Speech Production Model that comprises three main elements: the Conceptualizer, the Formulator and the Articulator. In the Conceptualizer, a preverbal message is generated by the speaker by drawing on his or her world knowledge, situational knowledge and linguistic knowledge. This preverbal message is passed on to the Formulator, where it is converted into a linguistically appropriate form. For this, the necessary lemmas are accessed in the speaker's mental lexicon and the preverbal message is encoded grammatically. The latter involves both the combining of the lemmas in a manner appropriate to the syntactic and semantic requirements of the speaker's language and the encoding of them into an appropriate phonological form. The resulting phonetic plan is

passed on to the Articulator, where it is realized as overt speech. All processes of speech production are assumed to occur with sequential overlap, i.e. while one (part of an) utterance is being articulated the following utterance is already encoded grammatically and phonologically in the Formulator and further utterances are being planned in the Conceptualizer. In addition, the entire speech production process is monitored by the speaker and can be interrupted or halted if necessary.

In practised speakers, the cognitive processes of speech production run automatically, i.e. without conscious effort or control, thus achieving the usually observable considerable speed of speech production. The process of speaking is thus conceptualised as forming part of a practised speaker's procedural knowledge, the largely subconscious knowledge of how to carry out actions. In contrast, less practised speakers such as language learners initially cannot draw on automatic speech production processes. For them, producing speech requires working memory capacity and conscious attention and thus forms part of the speaker's conscious declarative knowledge. It has been argued that all knowledge is initially declarative and that for behaviour that requires rapid performance such as fluent speech, declarative knowledge needs to be converted into procedural knowledge. The language learner's task is thus to develop procedural knowledge from declarative knowledge (Towell et al. 1996). The highly disfluent speech of some non-native speakers is assumed to reflect that the learner has not yet fully automatized the cognitive processes involved in speech production. Absolute fluency is reached when speech production requires neither attention nor effort, i.e. when a balance between automaticity and controlled processing has been achieved (Segalowitz 2000).

Based on this psycholinguistic definition of fluency, a number of temporal measurements of fluency have been developed. They measure quantifiable components of fluency such as pauses, speech rate and length of speech run. Two types of pauses can be distinguished: unfilled pauses that contain silence or non-speech acoustic events such as breathing and noise and filled pauses that contain nonlexical fillers such as “*uh*” and “*erm*” and elongations of

sounds (Riggenbach 1991, Freed 1995). Measurements of correlates of fluency in terms of pauses include the number and mean length of each type of pause (Lennon 1990, Freed 1995, Cucchiari et al. 2000, 2002), the pause ratio, defined as the total pause time as a percentage of the total time of speech (Lennon 1990, Temple 2000), and the number of filled or unfilled pauses per minute (Cucchiari et al. 2002) or per 100 syllables (Temple 2000).

Rate of speech has been measured in many different ways, for example as the number of words per minute (Lennon 1990), the number of syllables per minute (Towell 2002, Gut 2009, Shrosbree 2015) or the number of phonemes per time unit (Cucchiari et al. 2000, 2002, Fuchs 2016). Some researchers distinguish between speech rate, which relates the production of words, syllables and phonemes to the amount of speech produced by the speaker including pauses, and articulation rate, which measures the rate of words/syllables/phonemes in relation to the articulation time, thus excluding pauses of any kind (e.g. Gut 2009). Further, speakers' mean length of run (mlr) is measured as a correlate of fluency: for this, a 'run' is usually defined as an utterance, i.e. stretch of speech between pauses (Deschamps 1980, Lennon 1990, Freed 1995, Towell 2002, Cucchiari et al. 2000, 2002, Freed et al. 2004, Segalowitz & Freed 2004, Gut 2009) with some researchers including filled pauses in runs and others not. Since a run is defined by its delimitation by pauses of a certain length, it does not necessarily represent a semantic or syntactic unit in speech.

Gut (2009) tested the predictive ability of these quantitative measurements of fluency to distinguish reliably between native and non-native speech as well as their intercorrelation and the influence of speaking-style. For this, she analysed the phonologically annotated LeaP corpus (Gut 2014a), which contains 73,941 words produced by more than 130 native and non-native speakers of German and English. The following measurements were carried out:

- silent pause ratio (total duration of silent pauses divided by total duration of speech times 100)

- filled pause ratio (total duration of filled pauses divided by total duration of speech times 100)
- speech rate (total number of syllables divided by total duration of speech including pauses and non-speech events such as laughter)
- articulation rate (total number of syllables divided by total duration of speech excluding pauses and non-speech events)
- mean length of run (syllables per utterance)

Her results show that speech rate and articulation rate are highly correlated in both native and non-native speech, just as speech rate and mlr are. The filled pause ratio is also highly correlated with the three measurements speech rate, articulation rate and mlr in non-native speech. The silent pause ratio is not correlated with any of the other measurements in non-native speech, but it does not distinguish between native and non-native speech. Thus, only mean length of run and articulation rate are two reliable measurements of fluency that distinguish between native and non-native speech. Moreover, in native speech they do not correlate with each other. For both native and non-native speakers, mlr is higher when reading a scripted passage (where the processes of grammatical encoding of a preverbal message are not required because the text is given) compared to when retelling a story. We thus draw the tentative conclusion that the measurement mlr might reflect the speed of the processes in the Formulator, while articulation rate reflects the efficiency of the processes in the Articulator.

Fluency in NigE and ScotE might also be influenced by the sociolinguistic context of these varieties. The standard view, following Kachru (1985), is that varieties of English can be divided into those where English is used as a Native Language (ENL, e.g. ScotE), where English is used as a Second Language (ESL, e.g. NigE), and those where English is used as a Foreign Language (EFL, e.g. English in Germany). Much research has in the last decades argued that ESL differs from EFL, since ESL speakers usually rely (at least implicitly) on

locally emerging norms of English, while EFL speakers rely on norms from outside (usually British or American English, Kachru 1985, Schneider 2007). For example, NigE has developed extended meanings of *even* and *still* that are unknown in BrE (Fuchs et al. 2013). These innovations are locally meaningful and appropriate because Nigerians learn English first of all to communicate with other Nigerians. By contrast, innovations used by and known only to German learners of English might impede communication, since English is used in Germany mainly for communication with those who do not speak German.

While the distinction between ESL and EFL varieties is undoubtedly still useful, recent research has revealed that a strict divide between ESL and EFL in structural terms might be untenable in some cases. This is true for example for syntax and morphology (Edwards 2014, Edwards & Laporte 2015, Gilquin 2015, Gilquin & Granger 2011), for which Edwards (2014) showed that stative progressives occur both in ESL and EFL varieties. But it might also be true for phonology (Gut et al. 2015). For example, many ESL and EFL varieties have a tendency towards syllable-timing, while most ENL varieties tend towards a more stress-timed rhythm (Fuchs 2016: 87-102, Fuchs and Wunder 2015). However, there also appear to be phonological variables that set EFL varieties apart from ESL and ENL varieties. Analysing a range of fluency markers in three EFL, ESL and ENL varieties, Götz (2015) found that the EFL variety had the lowest degree of fluency, while there was no straightforward difference in fluency between ENL and ESL.

Thus, with regard to fluency in ScotE and NigE, previous research leads us to expect that NigE and ScotE speakers might differ in fluency only when a high degree of automatization is required to produce speech fluently. This is not the case for mean length of run (mlr) in scripted speech, which relies on the reading out of a script. We therefore expect a lower mlr for NigE in unscripted, but not in scripted speech. As far as articulation rate is concerned, we expect differences between the two speaker groups with a lower articulation

rate for the Nigerian speakers due to their lower degree of automatization of the articulation processes.

Method

We analysed scripted speech (taken from the ICE category broadcast talks; S2B-021 to 040) and unscripted speech (taken from the ICE category unscripted speeches; S2A-021 to S2A-050) from both ICE Nigeria and ICE Scotland. Table 1 shows the number of words, utterances and speakers in each text category and each ICE corpus.

Table 1. Number of words and speakers per text category that were analysed in ICE Nigeria and ICE Scotland.

	Scripted speech (broadcast talks)			Unscripted speech (unscripted speeches)		
	No. of words	No. of utterances	No. of speakers	No. of words	No. of utterances	No. of speakers
ICE Nigeria	9,708	2,127	10	4,488	1,119	8
ICE Scotland	8,813	1,520	18	4,229	896	6

All *ELAN* files were converted into Praat files (Boersma 2014; www.praat.org) and phonemic transcriptions were generated by using the phonemic forced alignment system MAUS (Munich Automatic Segmentation System, Reichel 2012, Schiel 2004). As described in section 2, MAUS maps the orthographic transcription onto a phonological transcription and time-aligns it with the sound recording. Figure 5 shows an extract from a broadcast talk from ICE Scotland that was phonemically aligned with MAUS and subsequently manually corrected in Praat. It shows, from top to bottom, the waveform, the spectrogram, the annotation tier for the orthographic transcription of words and the generated phonemic

transcription of both words and individual phonemes in SAMPA, a computer-readable variant of the IPA.

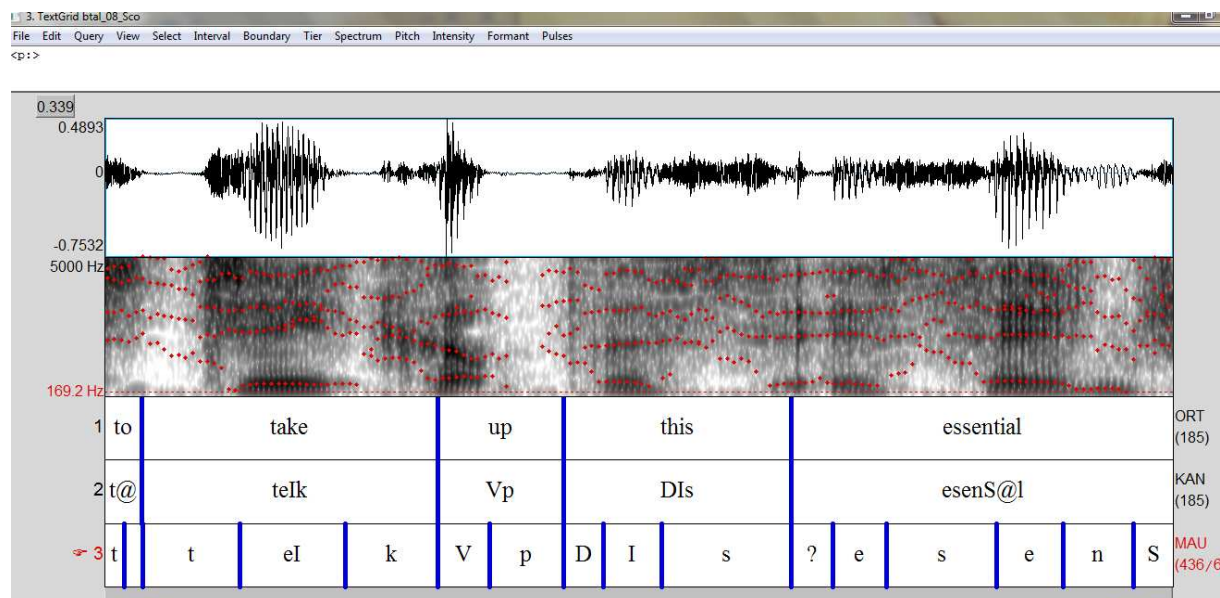


Figure 5. Manually corrected phonemic annotation of a broadcast talk in ICE Scotland. Extract from the phrase “to take up this essential role”.

All utterance boundaries and the number of phonemes per utterance were checked and, if necessary, corrected manually by two independent raters. In addition, all utterances that contained hesitations or disfluencies were excluded from analysis. Two measurements were taken with the help of a Praat script:

- the mean number of words per utterance (mlr) and
- the mean number of phonemes in total articulation time (articulation rate)

Mixed effects regression models (R function lme from package nlme, Pinheiro et al. 2015) with SPEAKER as random factor were calculated for ARTICULATION RATE and MLR per utterance. Significance testing was conducted with post-hoc Tukey tests after removing data points with a standardized residual of more than 2.5 standard deviations above or below the

mean to deal with the effect of extreme outliers (in effect less than 2.5 % of the data; R functions `glht` and `romr.fnc` from packages `multcomp` and `LMERConvenienceFunctions`, respectively; Hothorn et al. 2008, Tremblay et al. 2015).

Results

Figure 6 shows that in the scripted broadcast talks the Scottish speakers have a significantly higher MLR (5.8 words) than the Nigerian speakers (4.6 words, post-hoc test $z=2.4$, $p<0.05$, main effect for GROUP, $F=5.8$, $p<0.05$). In unscripted speeches, the Scottish speakers also have a higher MLR (4.8 words) than the Nigerian speakers (4.0 words), but the difference is smaller and not significant (post-hoc test $z=0.3$, n.s., no main effect for GROUP, $F=0.1$, n.s.).

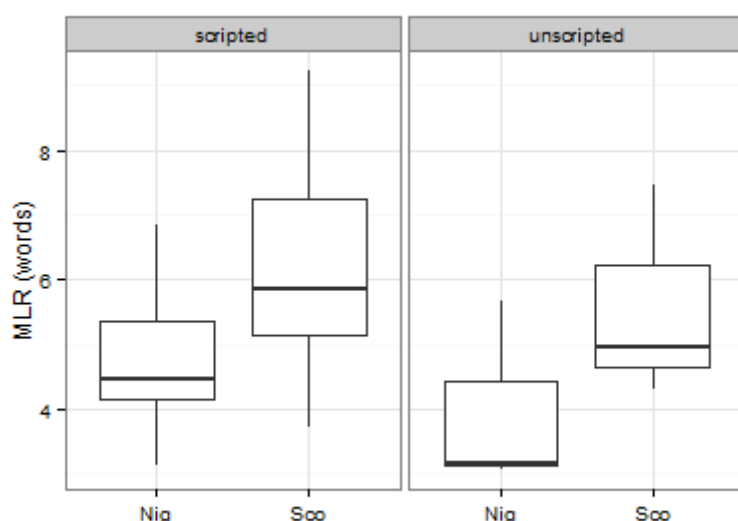


Figure 6. Mean length of run produced by the Nigerian and Scottish speakers in scripted speech (broadcast talks) and unscripted speech (unscripted speeches). Bold horizontal lines indicate medians, boxes correspond to the 25th and 75th percentiles, and notches to the 95th percentile.[§]

[§] This and the following figure were generated with the R function `ggplot` from package `ggplot2` (Wickham 2009).

Figure 7 shows that ARTICULATION RATE increases with UTTERANCE LENGTH for both speaker groups and in both speaking styles (significant main effect for UTTERANCE LENGTH, btal $F=581.1$, $p<0.001$, unsp $F=386.6$, $p<0.001$). For utterances shorter than 15-20 phonemes, ARTICULATION RATE rises sharply with increasing utterance length. For longer utterances, ARTICULATION RATE still depends on utterance length, but with only moderate increases.

When this is taken into account, Scottish speakers produce a higher overall ARTICULATION RATE in the scripted broadcast talks ($z=2.3$, $p<0.05$; main effect for GROUP $F=4.7$, $p<0.05$), while there is no significant difference between the two speaker groups in the unscripted speeches ($z=1.4$, n.s.; no main effect for GROUP $F=2.2$, n.s.). In addition, there are significant interactions between ARTICULATION RATE and GROUP in both scripted and unscripted speech (btal $F=30.2$, $p<0.001$; unsp $F=25.9$, $p<0.001$), such that articulation rate increases slightly more with increasing utterance length for Nigerian than for Scottish speakers.

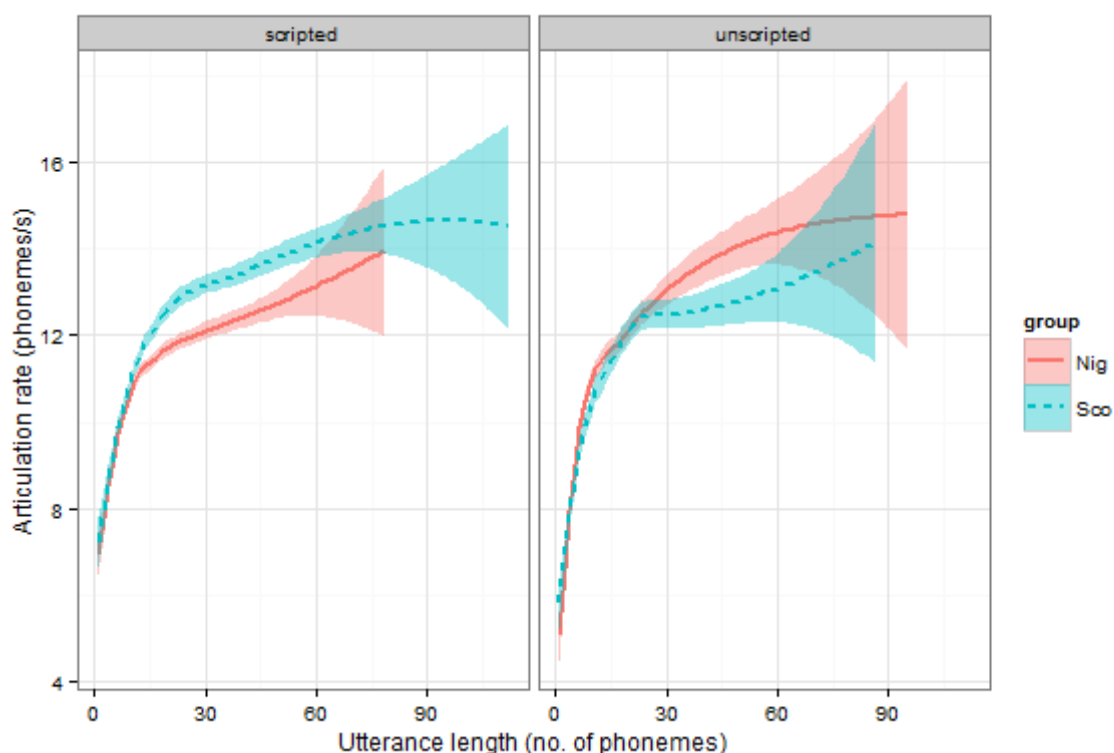


Figure 7. Articulation rate in relation to utterance length for the Nigerian (solid line) and Scottish speakers (broken line), with 95 % confidence intervals.

DISCUSSION

The results of this pilot study suggest that there might be differences in reading experience between the two groups with the Scottish speakers exhibiting a higher mean length of run in scripted speech compared to the Nigerian speakers. Furthermore, our hypothesis that differences in the speed of the Articulator processes might exist between ENL and ESL speakers of English was only partially confirmed, as articulation rate was significantly higher for Scottish speakers in scripted speech but not in unscripted speech.

Our study is the first to compare fluency in an ESL and an ENL variety while being both based on phonetic data and on a comparison of different speaking styles. Nevertheless, our findings can be compared to previous research. The analysis revealed a higher mlr for ScotE than for NigE in at least one speaking style, i.e. a higher mlr for ENL than for ESL. This confirms previous results by Götz (2015). However, mlr is also much higher across the board in her data, which is likely due to differences in methodology. Götz analysed orthographic transcriptions and measured mlr based on a definition of ‘run’ as “a speaker’s utterance until the occurrence of a disfluency” (2015: 1). This contrasts with our approach, which does not rely on previously existing transcriptions but rests on a phonological definition (utterances as intonation phrases).

Moreover, our pilot study suggests that the processes in the Articulator are not equally automatized for Nigerian and Scottish speakers of English, at least in scripted speech. Nigerian English speakers thus appear to differ from EFL speakers of English such as those analysed in Gut (2009), who had a lower articulation rate than ENL speakers regardless of speaking style. By contrast, the fact that the interaction between length of run and articulation rate (i.e. longer utterances are spoken at a faster rate) occurs for both the Nigerian and Scottish speakers analysed in the pilot study suggests that for both groups the interplay between Formulator and Articulator processes is the same and is arguably governed by ‘soft preplanning’ (Fery & Kügler 2008, Liberman & Pierrehumbert 1984) in both varieties. The

present pilot study thus contributes to the ongoing debate on differences between and similarities of ENL, ESL and EFL varieties in phonology (Fuchs & Wunder 2015, Götz 2015, Gut et al. 2015), lexis (Callies 2016, Edwards & Lange 2016) as well as morphology and syntax (Edwards 2014, Edwards & Laporte 2015, Gilquin 2015, Gilquin & Granger 2011, Koch et al. 2016, Meriläinen & Paulasto in press, Schneider & Gilquin 2016).

This paper has presented only one example of how the time-aligned and phonemically annotated ICE corpora of the new generation can be used to investigate a wide range of questions that were previously beyond our reach. They furthermore provide rich naturalistic data for research on the phonology of English varieties and allow studies on e.g. vowel and consonant inventories and realisations, intonation and stress patterns as well as speech rhythm. Likewise, research on discursal interactions between speakers by studying e.g. the length and frequency of their turns becomes possible. We look forward to further exciting innovative uses of the new generation ICE corpora.

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Appendix

Forced Alignment with MAUS

Note: This guide is based on the MAUS user interface as of September 25, 2015, and the interface might have changed since then.

1) Recommended but optional: Open eaf file in *ELAN* and delete all utterances with hesitations or false starts.

2) Convert *ELAN* annotation file into Praat annotation file: Choose File > Export As > Praat TextGrid. In the dialog, uncheck all boxes except the one for the annotation tier. If there are several speakers in the file, this and the following steps have to be repeated for each speaker individually (and the files can be merged afterwards).

3) Rename TextGrid tier from "Transcription 1" to "Transcription1" (either in Praat or by opening the TextGrid file with a text editor and searching for and replacing "Transcription 1").

4) Service 'Chunk preparation' at <https://clarin.phonetik.uni-muenchen.de/BASWebServices/#!/services/ChunkPreparation>

Upload TextGrid file.

Choose the following options (and leave the remaining fields unchanged):

- Language: English - Great Britain (or as appropriate)
- Input tier name: Transcription1
- Sampling rate: 16000

Choose "Run Web Service" and save the par file produced by this service.

5) Change bit rate of audio file: Drag and drop audio file into Audacity (<http://audacityteam.org/>), then at bottom left, change Project Rate (Hz) to 16000. Now choose File > Export, save as wav.

6) Service 'WebMAUS General' at <https://clarin.phonetik.uni-muenchen.de/BASWebServices/#!/services/WebMAUSGeneral>

Upload wav file under 'Signal-file-upload'

Upload par file from step 4 under 'Par-file-upload' (Note that the two files should have the same file name except for their extension).

Choose the following options (and leave the remaining fields unchanged):

- Language = English – Great Britain (or as appropriate)
- KAN tier in Textgrid = true
- ORT tier in Textgrid = true
- Chunk segmentation = true

Choose "Run Web Service" and save the TextGrid file produced by this service.